Concepts of current:

If dq is the amount of net charge that passes through any surface in a ~~thi~~ time interval dt, then the current 'i' is given by,

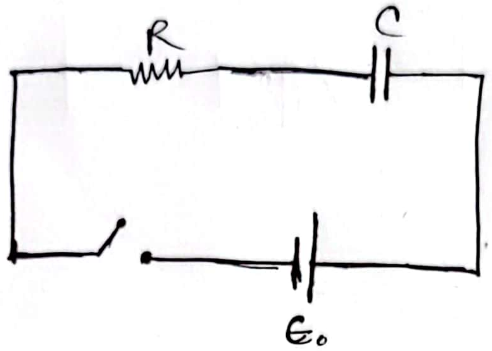
$$i = \frac{dq}{dt}$$

If the rate of flow of charge is constant in time, the current is said to be steady ~~or~~ or constant. Denoting the constant current by 'I' we simply get,

$$I = \frac{q}{t}$$

where q is the total ~~current~~ charge flow ~~of~~ in time t.

RC Circuit: The circuit which ~~of~~ is consist of Resistor & Capacitor is called R-C circuit.



Conditions—

i) Growth of a charge in capacitor (Charging).

ii) Decay of a charge in capacitor (Discharging)

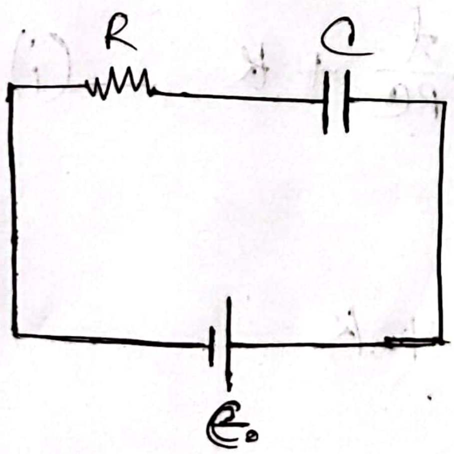
Growth of a charge:

The potential difference ~~accro~~ across the Resistor,

$$V = RI$$

And the potential difference across the Capacitor,

$$V = \frac{q}{C}$$



According to KVL,

$$\mathcal{E}_0 = RI + \frac{q}{C}$$

$$\Rightarrow \mathcal{E}_0 = R \cdot \frac{dq}{dt} + \frac{q}{C}$$

$$\Rightarrow \mathcal{E}_0 - \frac{q}{C} = R \cdot \frac{dq}{dt}$$

$$\Rightarrow \frac{\mathcal{E}_0 C - q}{C} = R \cdot \frac{dq}{dt}$$

$$\Rightarrow \frac{C}{\mathcal{E}_0 C - q} = \frac{dt}{R \cdot dq}$$

$$\Rightarrow \frac{dq}{\mathcal{E}_0 C - q} = \frac{dt}{RC}$$

$$\Rightarrow \frac{dq}{q - \mathcal{E}_0 C} = -\frac{dt}{RC}$$

$$\Rightarrow \int \frac{dq}{q - \mathcal{E}_0 C} = -\int \frac{dt}{RC}$$

$$\Rightarrow \ln|q - \mathcal{E}_0 C| = -\frac{t}{RC} + K \quad \text{--- (i)}$$

when $t = 0$, then $q = 0$,

$$\ln|0 - C\mathcal{E}_0| = -0 + K$$

$$\therefore K = \ln|-C\mathcal{E}_0|$$

placing value of K in eqn (i),

$$\ln|q - C\varepsilon_0| = -\frac{t}{RC} + \ln|-C\varepsilon_0|$$

$$\Rightarrow \ln\left|\frac{q - C\varepsilon_0}{-C\varepsilon_0}\right| = -\frac{t}{RC}$$

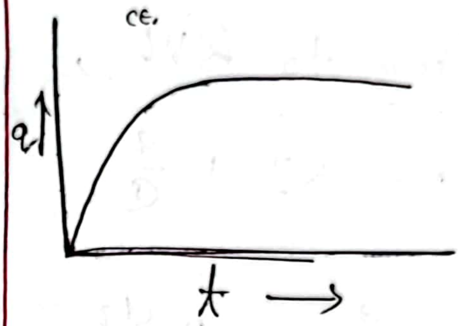
$$\Rightarrow -\frac{q}{C\varepsilon_0} + 1 = e^{-t/RC}$$

$$\Rightarrow \frac{q}{C\varepsilon_0} - 1 = -e^{-t/RC}$$

$$\Rightarrow \frac{q}{C\varepsilon_0} = 1 - e^{-t/RC}$$

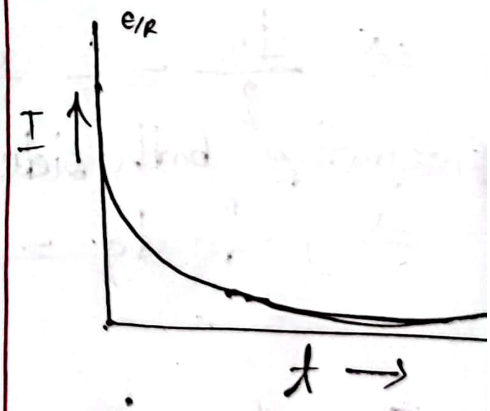
$$\boxed{\therefore\ q = C\varepsilon_0\left(1 - e^{-t/RC}\right)}$$

Graph:



$\boxed{\text{For maximum charge, } q_0 = C\varepsilon_0}$

Graph:



we know,

$$I = \frac{dq}{dt}$$

$$= \frac{d}{dt}\left\{C\varepsilon_0\left(1 - e^{-t/RC}\right)\right\}$$

$$= C\varepsilon_0 \cdot \frac{d}{dt}\left(1 - e^{-t/RC}\right)$$

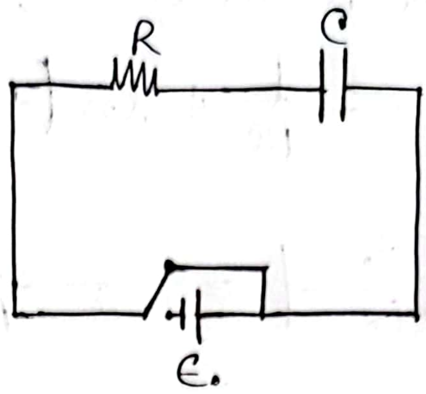
$$= C\varepsilon_0 \cdot \left(0 + e^{-t/RC} \cdot \frac{1}{RC}\right)$$

$$= C\varepsilon_0 \cdot \frac{1}{RC} \cdot e^{-t/RC}$$

$$\boxed{\therefore\ I = \frac{\varepsilon_0}{R} \cdot e^{-t/RC}}$$

CamScanner

Decay of a charge:



Again,

$V_R = RI$

and, $V_C = \frac{q}{C}$

According to KVL,

$$0 = RI + \frac{q}{C}$$

$$\Rightarrow -\frac{q}{C} = R\cdot\frac{dq}{dt}$$

$$\Rightarrow -\frac{C}{q} = \frac{dt}{R\cdot dq}$$

$$\Rightarrow -\frac{dq}{q} = \frac{dt}{RC}$$

$$\Rightarrow \frac{dq}{q} = -\frac{dt}{RC}$$

Integrating both side,

$$\Rightarrow \int \frac{1}{q}\, dq = -\frac{1}{RC}\int dt$$

$$\Rightarrow \ln|q| = -\frac{t}{RC} + K \quad \text{———} \quad (i)$$

when, $t = 0$, then, $q = q_0 = CE_0$

$$\Rightarrow \ln|q_0| = 0 + K$$

$$\therefore K = \ln|CE_0|$$

placing the value of K in eqn (i),

$$\ln|q| = -\frac{t}{RC} + \ln|C\mathcal{E}_0|$$

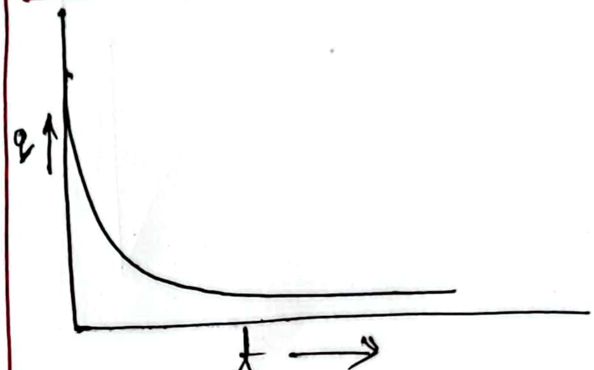
$$\Rightarrow \ln|q| - \ln|C\mathcal{E}_0| = -\frac{t}{RC}$$

$$\Rightarrow \ln\left|\frac{q}{C\mathcal{E}_0}\right| = -\frac{t}{RC}$$

$$\Rightarrow \frac{q}{C\mathcal{E}_0} = e^{-t/RC}$$

$$\boxed{\therefore q = C\mathcal{E}_0 \cdot e^{-t/RC}}$$

Graph:



we know,

$$I = \frac{dq}{dt}$$

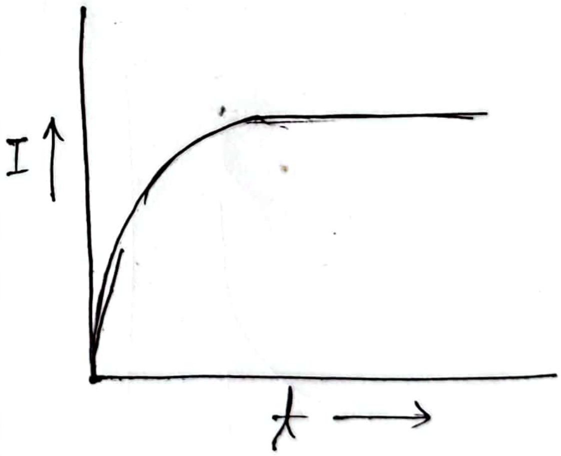
$$= \frac{d}{dt}\left(C\mathcal{E}_0 \cdot e^{-t/RC}\right)$$

$$= C\mathcal{E}_0 \frac{d}{dt}\left(e^{-t/RC}\right)$$

$$= C\mathcal{E}_0 \cdot e^{-t/RC} \cdot \left(-\frac{1}{RC}\right)$$

$$= -\frac{\mathcal{E}}{R} e^{-t/RC}$$

Graph:



(*) Max. current, $I_0 = \frac{\mathcal{E}}{R}$